Intermolecular Forces Lab

Chem 151

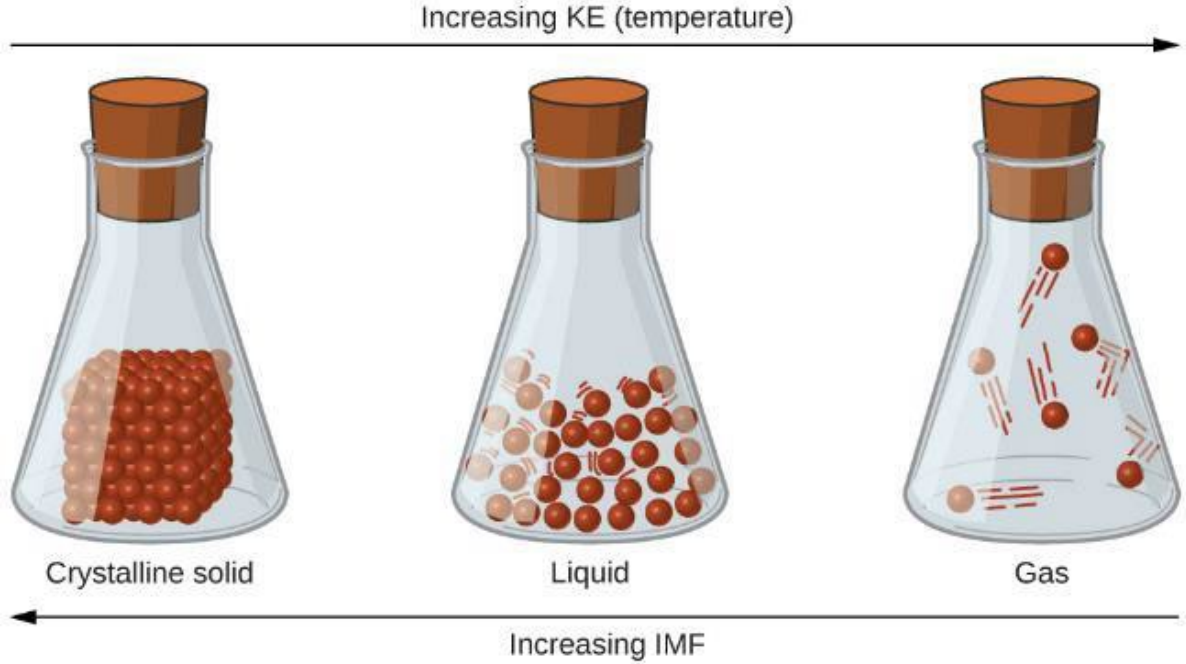
Sample of Reaction You are Studying:

n-pentane(l) → n-pentane(g)

Molecules of pentane are spreading out

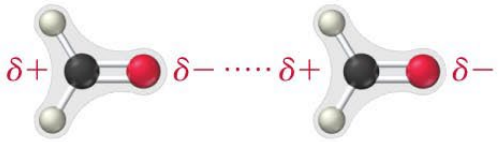
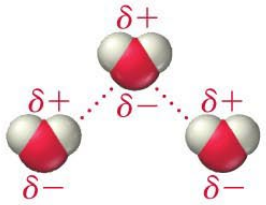
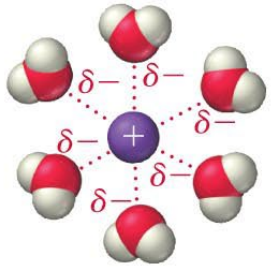
You are breaking intermolecular bonds between pentane molecules

Breaking bonds requires energy – measure relative energy with ΔT



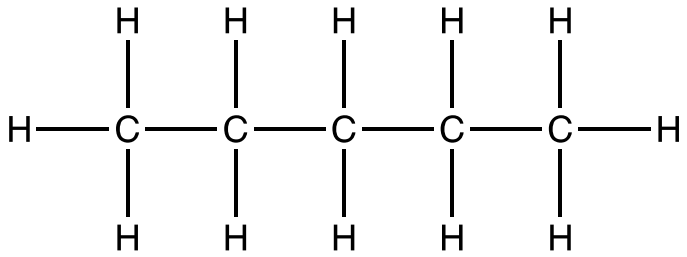
Types of Intermolecular Forces

TABLE 11.4 Types of Intermolecular Forces

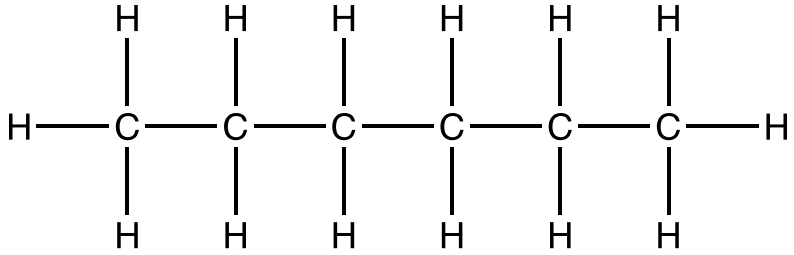
Type	Present in	Molecular perspective	Strength
Dispersion	All molecules and atoms		
Dipole–dipole	Polar molecules		
Hydrogen bonding	Molecules containing H bonded to F, O, or N		
Ion–dipole	Mixtures of ionic compounds and polar compounds		

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Structural Formulas of Molecules of Interest Today:

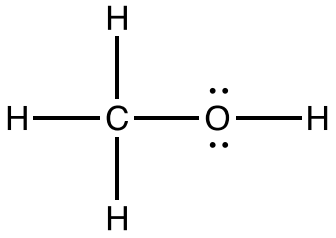


n-pentane

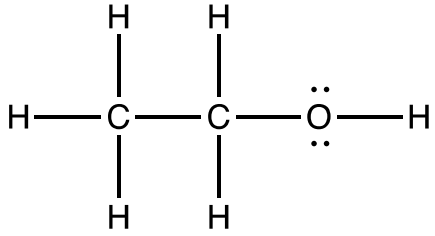


n-hexane

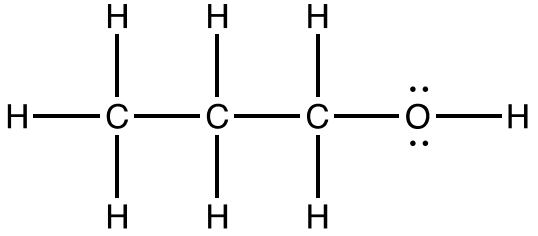
What types of intermolecular forces are present?



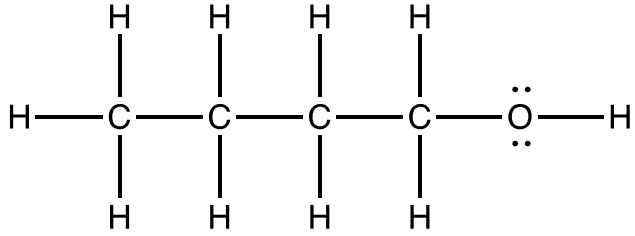
methanol



ethanol



propanol



butanol

Table 1. Pre-Lab Table

Substance	**Formula**	**Structural Formulas**	**Molar Mass**	**H-Bond?** (Yes or No)
ethanol	C_2H_5OH			
1-propanol	C_3H_7OH			
1-butanol	C_4H_9OH			
n-pentane	C_5H_{12}			
methanol	CH_3OH			
n-hexane	C_6H_{14}			

Calculations:

Example: Molar Mass of ethanol (C_2H_5OH) = 2 Carbon 2 x 12.01 amu

6 Hydrogen 6 x 1.008 amu

1 Oxygen 1 x 15.999 amu

46.068 amu = **46.068 g/mol**

Experimental Set-Up

